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1. Introduction

The objective of this coursework is to design a small-scale structure idealised to a single-degree-of-freedom system (Section 2.1). It must be able to withstand three different dynamic loads applied in the form of a periodic base motion with displacement amplitudes of 10mm, 10mm and 20mm and frequencies of 1.5Hz, 2.5Hz and 3.5Hz respectively.

The structure is to be 3D printed with ABS or PLA materials. The maximum tensile or shear stress experienced by the structure under the dynamic loads should not exceed the strength limits to prevent breaking. These requirements are analysed in Sections 3.2 to 3.4. In addition, the design choices are to be made taking in consideration the elegance of the structure, as discussed in Section 3.5. The optimum structure is one that satisfies all the structural and aesthetic requirements, while minimising the total mass.

2. Drawing

2.1 Final Design Dimensions

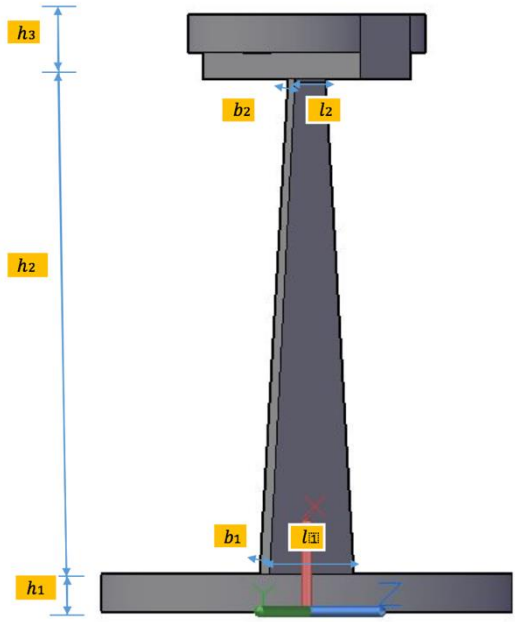


Figure 2.1: 3D structure that we designed

The final dimensions of our structure are shown in the table below,

Component of the Model	Dimension (mm)
Total Height, H	160
Base Thickness, h_1	10
Top Plate Thickness, h_3	10 + 7
Column Height, h_2	133
Column Width (Top)	15
Column Breath (Top), b_2	2
Column Width (Bottom), l_1	42
Column Breadth (Bottom), b_1	3
Diameter of Holes on Base Plate, d_1	10
Diameter of Holes on Top Plate, d_2	7

Table 2.1: Dimensions of the structure

2.2 Printing Axes

We are using Fused Filament Fabrication (FFF) process to 3D print our model, which is a continuous printing process using heated thermoplastic material. During the process, the head moves in 2D to deposit one horizontal layer at one time, before moving along the z axis to print another new layer. To print the model efficiently, we decided to orientate our model along the x, y and z axes as shown in the Figure 2.

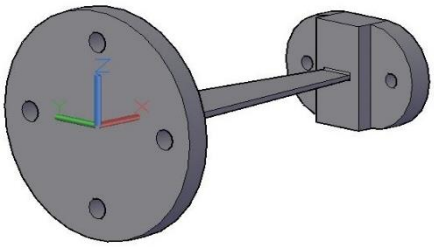


Figure 2.2: Printing axes

The top and the base plates of the model are to be printed with 50% infill, while the column (key) element is to be printed solid with no infill. To make our structure light and efficient, the model is printed in an orientation such that it keeps the amount of solid with no infill in both the top and base plates to minimum. This orientation also ensures the printed model has an overall higher tensile strength

3. Design rationale and calculations

3.1 Formulation of the equation of motion

Total acceleration of mass: $\ddot{u}_t(\mathrm{t}) = \ddot{u}_g(\mathrm{t}) + \ddot{u}_{st}(\mathrm{t})$

Ground acceleration:
$u_g(\mathrm{t}) = A\sin(\omega t)$
$\dot{u}_g(\mathrm{t}) = A\omega\cos(\omega t)$
$\ddot{u}_g(\mathrm{t}) = -A\omega^2\sin(\omega t)$

Newton's second law:
$-f_I(t) - f_D(t) - f_R(t) = 0$
$-m\ddot{u}_t(\mathrm{t}) - c\dot{u}_{st}(\mathrm{t}) - ku_{st}(\mathrm{t}) = 0$
$-m\ddot{u}_g(\mathrm{t}) - m\ddot{u}_{st}(\mathrm{t}) - c\dot{u}_{st}(\mathrm{t}) - ku_{st}(t) = 0$
$\ddot{u}_{st}(\mathrm{t}) + \frac{c}{m}\dot{u}_{st}(\mathrm{t}) + \frac{k}{m}u_{st}(\mathrm{t}) = -A\omega^2 sin\omega t$

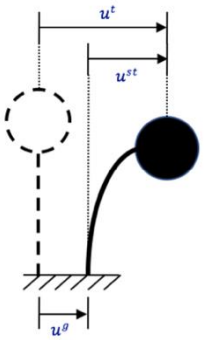


Figure 3.1.1: 1 DoF model

Whereby: u_g = ground displacement, u_{st} = dynamic displacement, u_t = total displacement, f_I = inertia force, f_R = resistance force from column, f_D = damping force, k = stiffness, c = damping coefficient, A = amplitude.

3.2 Design of top platform

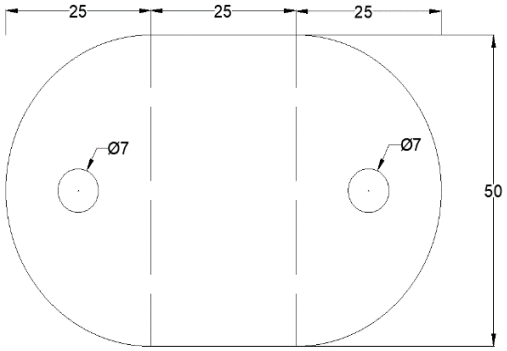


Figure 3.2.1: Plan view of top

As specified in the coursework brief, clear flat space of 50 mm by 25 mm should be provided at the top of the structure to place measuring equipment. The forcing direction is parallel to the 50mm breadth.

Additional masses are to be fixed by nuts and bolts of 7mm diameter through the holes on both ends of platform. We design the additional space to be semicircle, allowing us to save on the edge material. The holes are designed to be in the middle of the additional space, allowing a 9mm edge-to-edge distance to prevent end-tearing.

The structure is simplified as a 2-noded cantilever system, which will be explained in Section 3.3.3. The length of the cantilever is required in the calculation of the system's stiffness, where it is measured from the fixed end to the centre of mass of the top half of the structure (including two 250g masses). To avoid complication, the thickness of the top platform is decided to be 17mm as shown in Figure 3.2.2 such that the centre of mass lies at the top end of the cantilever column. An example of complication is the value of second moment of area that should be used if the centre of mass lies above the top end of the column.

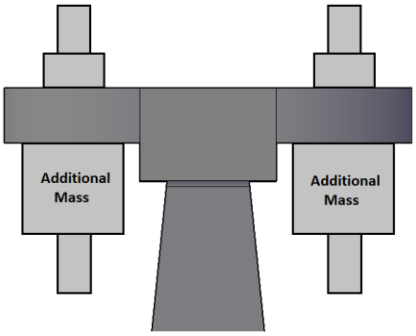


Figure 3.2.2: Side view of the top platform

The calculations are shown in Appendix B. The remaining area of the top platform is designed to be 10mm.

3.3 Design of cantilever column

3.3.1 Selection of column shape

When designing the column, there were several cross-sectional shapes to choose from. Namely, circular, square or a rectangular column cross section. The most efficient shape in providing tensile resistance against dynamic loading was chosen. As tensile stress in the column is calculated by $\sigma_x = \frac{My}{I}$, this can be reduced by minimising the ratio of $\frac{y}{I}$ and maximising the second moment of area, I. Thus, a rectangular cross-section with its length (y-axis) and breadth (z-axis) would provide the largest I for the same cross-sectional area.

The column has also been designed to be tapered, with the column cross-section decreasing as z increases. This is to optimise the material efficiency of the structure. Moreover, as bending moments experienced by the structure

is largest at the fixed-end, the cross-sectional area at the support should hence be largest to resist a larger tensile axial stress.

3.3.2 Utilisation of MATLAB to obtain the optimal dimensions

A MATLAB script (included as Appendix A) was written to iterate through different dimensions of the column while maintaining the design of the top platform as set out in Section 3.2. For each combination of dimensions, the structural as well as strength requirements mentioned in Section 1.0 were checked to ensure that they were satisfied. The calculations for the chosen dimensions and the assumptions made are explained in the following sections. Printing errors have also been accounted for in the script, simulated as combinations of increments or decrements in the column dimensions by 0.1mm. This ensures that even if printing errors are present, the requirements are still satisfied.

3.3.3 Determination of natural period

The natural period of the structure is required to be bounded between 0.35s and 0.7s. The given limits restrict the choice of design structure as natural period depends on mass and stiffness. It is also important to note the possible occurrence of resonance as the excitation frequency approaches the natural frequency. The large amplitude oscillations due to the storage of vibrational energy by resonance should not exceed the maximum relative dynamic displacement of 3 cm. The following assumptions and methodologies are employed in calculating the natural period of the structure:

(i) Cantilever support condition

As the whole structure will be printed continuously, the column and base plate will be monolithically connected. Therefore, the base connection can be assumed to be a fixed connection. The structure will hence be modelled as a cantilever, with the length measured to the centre of mass.

(ii) Lumped mass assumption

The structure is modelled as a 2-noded structure, specifically at the top and bottom platforms. The self-weight of the structure is a uniformly distributed load, but the lumped mass assumption assumes that the column mass is transferred equally to the two nodes, as shown in Figure 3.3.1.

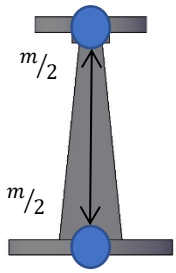


Figure 3.3.1: Lumped mass model

(iii) Formulate stiffness using Virtual Work

Due to our tapered column shape, the cross-section and thus the second moment of area will vary with the height of the column. Therefore, the column stiffness was derived using the Principle of Virtual Forces (PVF), specifically the unit load method which states that $\delta^1(U^*) = \delta^1(W^*)$. By applying a unit load at the free-end of the cantilever, we can obtain the deformations at that location, and using Hooke's Law, the stiffness of the cantilever is obtained as follows:

$$k = \frac{E}{\int_0^L \frac{(L-x)^2}{I(x)} dx} = 75.10\ N/m$$

$$\omega_n = \sqrt{k/m} = 11.96\ rad/s$$

whereby $m = 0.52\ kg$, sum of the mass of top half of structure and the two attached metal masses.

$$\omega_n = \frac{2\pi}{T_n}$$

Thus, the natural period of the structure is given by $T_n = 0.525\ s$.

3.3.4 Determination of the maximum relative dynamic displacement

The displacement of the structure will be induced by ground motion, as defined in Section 2.1. In order to determine the maximum dynamic displacement, u_{max} during each load case, the maximum static amplitude, u_{st} of the system and the dynamic amplification factor, R have to be determined:

$$u_{max} = u_{st} \cdot R$$

Where $u_{st} = \frac{F_0}{k}$ & $R = \frac{1}{1-\left(\frac{\omega}{\omega_n}\right)^2}$. Hence, taking $F_0 = m\ddot{u}_{g,max} = mA\omega^2$,

$$\therefore u_{st} = \frac{A\omega^2}{{\omega_n}^2} \quad \rightarrow \quad u_{max} = \frac{A\omega^2}{{\omega_n}^2 - \omega^2}$$

The expected u_{max} values are given in the following table:

	Case 1	**Case 2**	**Case 3**
A [mm]	10	10	20
f [Hz]	1.5	2.5	3.5
ω [rad/s]	3π	5π	7π
$\boldsymbol{u_{max}}$ **[mm]**	**16**	**24**	**28**

Table 3.3.1: u_{max} for each of the loading cases

3.3.5 Strength limits

It is common practise to determine the maximum force exerted on the structure based on the maximum acceleration of the ground. Since the displacement amplitude and the forcing frequency for Test 3 are the largest among the tests, the acceleration is expected to be the largest as calculated below:

$$F_{max} = F_0 = mf_0 = mA\omega^2 = 5.07N$$

The shear force diagram and the bending moment diagram can then be plotted considering the force acting on the free end of the cantilever column. The maximum shear force is equal to the maximum force applied ($V_{max} = F_{max} = 5.07N$) and the maximum bending moment at the fixed end is equal to the maximum force multiplied by the length of the column, $M_{max} = F_{max}\, h_2 = = 674Nmm$.

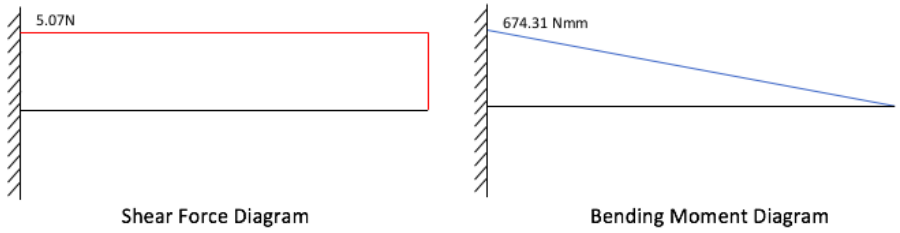


Figure 3.3.2: Shear force and bending moment diagrams.

Tensile strength

As mentioned in section 3.3.1, the tensile stress can be found from the equation in Section 3.3.1. However, M, y and I vary along the height of the column. By finding the independent equations for M, y and I, the tensile stress along the column can be found.

$$\sigma_z(z) = \frac{6M(z)}{l(z)b^2(z)}$$

From the above equation, an expression for σ_z is obtained. The value of σ_z is found for each 1mm in the column our MATLAB script, and a Factor of Safety of 1.3 was applied when comparing this to the tensile strength of the material. The FoS was applied to account for construction errors and material imperfection, as well as to make up for the negligence of additional moment induced when the structure is displaced.

Shear strength

The shear strength of the column also needs to be checked. Since the column behaves as a cantilever structure, the shear force is constant along the column height. The top section of the column at the connection to top platform is the smallest area, therefore generates the largest shear stress. This τ_{max} is checked against the allowable shear strength which is $\frac{1}{\sqrt{3}}$ of the allowable tensile strength.

$$\tau_{max} = \frac{V_{max}}{l_2 b_2} = 0.169N/mm^2$$

$\tau_{yield} = 8.66N/mm^2$, therefore $FoS = \frac{\tau_{yield}}{\tau_{max}} = 51.2 > 1$

Increase the area at the top end

The connection between the top end of the column to the top platform is susceptible to snapping since the column is very thin and the corner has an angle of less than 90 degree. Therefore, the corner is rounded.

3.3.6 Check for Euler buckling

Checks were carried out to ensure that the column does not fail under Euler buckling by comparing the weight applied on the column to the critical buckling load $\mathrm{P_{cr}}$. In a uniform cross-sectional column, $\mathrm{P_{cr}} = \frac{\pi^2 \mathrm{EI}}{\mathrm{L_e^2}}$. However, in a tapered column like ours, the second moment of area I varies with height. Therefore, the critical Euler buckling load had to be derived from the potential energy of the column using Rayleigh's method. The deflected shape of the column was modelled after one with pin-ended supports:

$$w = Qsin\left(\frac{\pi x}{L_e}\right)$$

where $L_e = 2h$ (the effective length for a cantilever)

$$V = \int_0^L \left[\frac{1}{2}EI(x)\left(\frac{d^2w}{dx^2}\right)^2 - \frac{1}{2}P\left(\frac{dw}{dx}\right)^2\right]dx$$

The system will become unstable when $\frac{d^2V}{dQ^2} = 0$, the bifurcation point. By solving this equation, the following expression for P_{crit} is obtained and evaluated with numerical integration in MATLAB:

$$P_{crit} = \frac{E\pi^2}{4L^2}\frac{\int_0^L I(x)sin^2\left(\frac{\pi x}{2L}\right)dx}{\int_0^L cos^2\left(\frac{\pi x}{2L}\right)dx} = 8.78kN$$

The force exerted by the mass on the top half of the structure of 5.14kN does not exceed this P_{crit} value.

3.4 Design of base plate

The base plate was designed to be 110mm in diameter and 10mm thick. During testing, the structure is to be fixed to the shaking table by nuts and bolts of 10mm diameter through the holes on the table. As the holes are spaced at 40mm intervals (centre-to-centre) in both directions from the centre of the table, the radius of the base plate had to be designed in accordance to these spacings. Moreover, to prevent tearing or fracturing of the base plate at the connections with the nuts and bolts, an edge distance of 10mm was added from the edge of the hole. As such, we obtained a base plate of length 110mm (illustrated in figure 3.4.1). Furthermore, a circular base plate was selected, allowing us to be more materially efficient by saving on edge material.

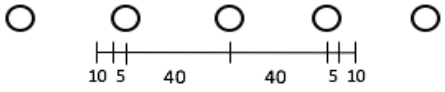


Figure 3.4.1: Shaking table hole spacing.

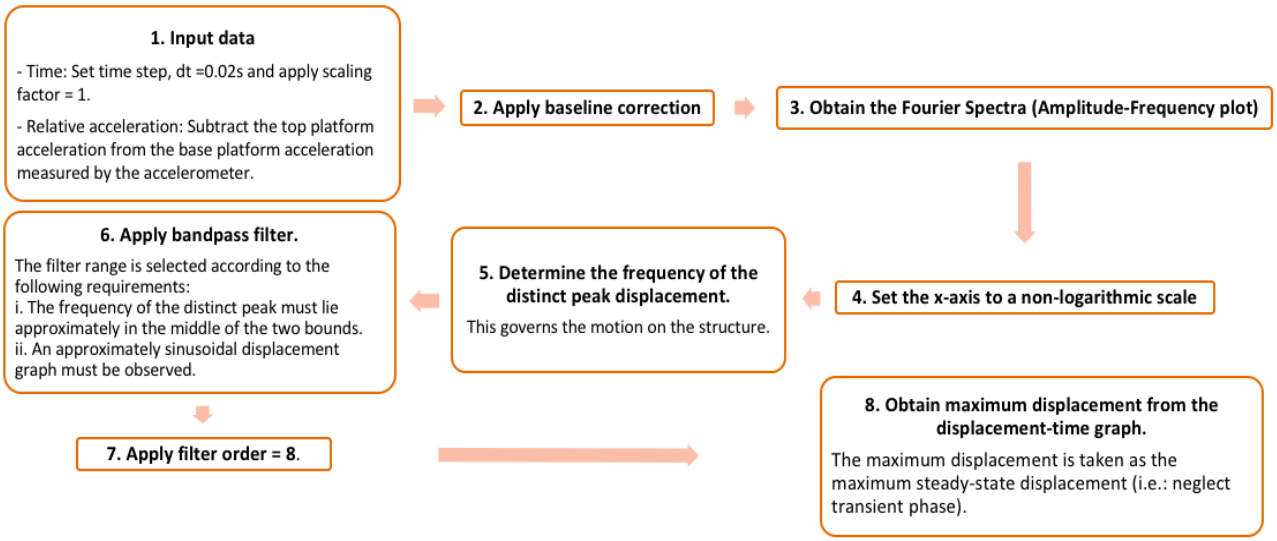
3.5 Aesthetic considerations

Works of structural engineering designs can be regarded as 'Structural Art' when they satisfy the *3 Es – Efficiency, Economy and Elegance.* (Billington, 1983). Elegant structures are often light weight structures (The Structural Engineer, 2015) with reduced visual complexity. The structure that we designed is light weight, simple and works efficiently at a minimum amount of material. The design decisions are highly informed by engineering considerations. These include the usage of tapered rectangular column, as well as the thickness of top platform such that the centre of mass lies at the top end of the column. As for economy, the structure is designed in a way to minimise material usage by choosing 50% infill at the top and bottom plates. It requires minimal amount of material to achieve a satisfactory service state.

4. Experimental results

4.1 Post-processing of Data

The post-processing of the data obtained by the accelerometer readings were carried out using SeismoSignal according to the procedure described in the above flow chart. The analysis results for all 4 loading cases are presented in tables and graphs attached in the Appendix C.



4.2 Discrepancy in natural period

The actual period that the system oscillates at under free vibration is 0.51s, which is smaller than the predicted natural period of 0.525s; the frequencies are 12.32rad/s and 11.96rad/s respectively. The four main factors are reduction in mass, increase in stiffness, presence of cable and negligence of damping.

If the structure is perfectly printed, the mass is estimated to be 78.81g. However, the mass of the structure is measured as 60g (rounded to the nearest 5g). The significant difference could be due to the top and bottom platforms are designed to be printed with 50% infill; this level of accuracy is difficult to achieve in 3D printing. Secondly, there could be errors in printing where the dimensions are smaller than what was designed for. Besides, the given density of ABS material could be overestimated since there could be slight variation between manufacturers.

Apart from the mass, the natural period is also dependent on the stiffness of the system. For a cantilever column with uniform cross-sectional area, the stiffness is given as $3EI/L^3$ (i.e. it is inversely proportional to the length cubed), implying that it is highly sensitive to the length of the cantilever column to the centre of mass. Since the actual frequency, thus the actual stiffness, is larger than predicted, it is inferred that the actual length of the cantilever column is shorter. During the design stage, the mass of the column is neglected in the calculation of the centre of mass (Appendix B) and the mass of the top platform is overestimated.



Figure 4.2.1: Small gap between masses and top

In addition, as shown in Figure 4.2.1, the measuring equipment obstruct the attachment of the masses onto the top platform, resulting in a small gap. All these contribute to a lower centre of mass. Furthermore, the cable that is connected to the measuring equipment is relatively stiff. This provides additional resistance to the system that might affect the natural period.

Lastly, it should be noted that the system oscillates at the damped frequency under free vibration; damped frequency, ω_d is always smaller than the natural frequency ω_n, where they are related by the damping ratio ξ:

$$\omega_d = \omega_n\sqrt{1-\xi^2}$$

The effects of damping are explained in more details in Section 4.3. However, the percentage difference between the observed and predicted periods of 2.86% is minor. It still lies within the required range between 0.35s and 0.7s.

4.3 Discrepancy in maximum displacements

In estimating u_{max} during the design phase, we neglected damping effects for more conservative estimates. Large differences were observed for the maximum displacement compared to what was predicted, as shown in Table 4.2.1.

	Predicted u_{max}	**Actual u_{max}**	**Percentage Difference**
Case 1	16	6.00mm	-62.5%
Case 2	24	26.02mm	8.4%
Case 3	14	6.23mm	-55.5%
Case 4	28	13.82mm	-50.6%

Table 4.2.1: Comparison of u_{max} values

Comparing test 3 and 4, the actual u_{max} are in-line with the predicted u_{max} trend, where u_{max} for case 4 is approximately double the value for u_{max} from case 3. This is because for both case 3 and 4, the structure vibrates at a frequency close to 3.5Hz, but the displacement amplitude for case 4 is double that of case 3's.

The discrepancy between the test values and calculated value can be explained using damping. While it was neglected in our calculations, in reality, this was present due to some degree of internal structural damping, the sources of which are as follows. Firstly, the material was assumed to be uniform, however in reality, discontinuity will exist on a microscale due to air bubbles and imperfections caused from 3D printing. Secondly, the foundation of the structure (bottom plate) was not completely solid as we initially designed for, due to it being printed 50% infill, which would result in the structure dissipating more energy. Finally, we assumed that the structure would be firmly fixed onto the base, however the nuts may not be completely tightened. In such cases, damping would occur with a damping coefficient estimated around 5-10%. By taking damping into consideration, the dynamic amplification factor is given by:

$$R = \frac{1}{\sqrt{\left(1-\left(\frac{\omega}{\omega_n}\right)^2\right)^2+\left(2\xi\frac{\omega}{\omega_n}\right)^2}}$$

where ξ is the damping ratio, instead of the equation utilised in Section 3.3.4. Therefore, R would be affected by the damping ratio, which would give us a different value for the predicted u_{max}.

Additionally, the natural period of 0.51s measured from free vibration is suspected to be a damped period, T_d. The actual natural period, T_n is therefore expected to be smaller than that, indicating that the natural frequency, f_n, of our structure is closer to 2.5 Hz (Case 2). Thus, this explains why the greatest displacement occurs in Case 2 compared to the rest of the loading cases. It is also worth noting that the actual frequency (1.66Hz, 2.57Hz and 3.61Hz) experienced by the structure during tests are greater than the initial specified value of 1.5Hz, 2.5Hz, 3.5Hz.

Another reason for the large discrepancies in values, was that our structure was not a single degree-of-freedom structure, as we assumed. From the videos provided, it was observed that our structure had at least 2 degrees-of-freedoms (dof), observed as a rotation of the top platform about the y-axis. This would have largely affected the actual results, resulting in the discrepancies with our predicted values.

However, as damping only largely affects R and u_{max} when $\frac{\omega}{\omega_n} \approx 1$ (close to resonance), these discrepancies could have been due to a combination of other factors, such as errors in the data provided. For example, when extracting the displacement values of the shaking table from the accelerometer readings, the amplitude obtained was 3mm and 6mm, instead of 10mm and 20mm respectively. This could mean that the accelerometer readings provided was smaller than it should have been.

4.4 Meeting strength constraints

The structure did not fail by material yielding, implying that the assumptions and the calculations made were safe. The maximum displacement observed in Test 2 of 26.02mm is much lower than the maximum static displacement of 67.52mm used in design. Besides, the applied factor of safety of 1.3 and 51.2 on tensile and shear stress respectively allows additional deviation, for example from eccentricity.

5. Critical analysis

Through this exercise, we were able to gain first-hand experience of the knowledge we learnt through its application to a 1 dof earthquake resistant structure. We were also able to gain appreciation of idealised equations. The system was modelled as 1 dof, however this is a great simplification as in reality there are many dofs. Whereas small effects were ignored through the use of linear second order differential equations, the actual design is much more complicated and requires non-linear analysis. Being able to accurately predict the damping of the system is also a matter of experience and expertise which comes with years of work experience. Therefore, theoretical equations alone are not enough to accurately predict the system behaviour; the problem can be better addressed using the widely used finite element method. Finally, compromise is necessary in order to balance time, cost and effort. The most elegant design will not always meet design safety requirements. Thus, the final design will be a compromise of all the factors above.

To improve the performance of our structure and the accuracy of the results, several changes can be implemented during the design stage, which are:

(i) Damping is one of the factors that causes discrepancies between our predicted maximum displacements and actual maximum displacements. Hence, it should be considered in the design stage.
(ii) As mentioned previously, the structure was designed based on 1-dof and the idealised equations simplified our design calculations. To improve the performance, we should consider the structure as multi degree-of-freedom.
(iii) Column mass has been neglected and lump mass assumption has been adopted in our design. A better way of distributing and calculating the mass should be done to improve the accuracy.
(iv) The printed length and mass of the structure are lower than the estimated one. For example, the estimated mass exceeded the printing mass by about 20%. To make our structure more efficient, these printing errors should be accounted during the design stage.

6. Conclusion

In conclusion, the cantilever structure was chosen as it was an elegant and efficient structure that satisfied the design constraints, and safety requirements. The linear second order differential equations utilised were sufficient to calculate an estimate of the structure's dynamic response. However, in reality, the structure responds non-linearly and therefore, the linear solutions will only provide a rough estimate of the response. Therefore, it is practical to utilise a more conservative framework when estimating the response, such as assuming undamped conditions and implementing Factors of Safety in the strength calculations.

Furthermore, it was unrealistic to oversimplify the structure to have only one degree of freedom. In reality, our structure had two dofs. Therefore, discrepancies with the experimental data were to be expected.

7. References

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Appendix A: Calculations

Centre of mass

The thickness of the top platform is designed such that the centre of mass lies at the top end of the cantilever column. It is assumed that the mass of the column and the holes is negligible, and that the centre of mass of the bolt and nut is 10mm from the top surface of the platform. This results in a quadratic equation of thickness t, which is derived and solved below:

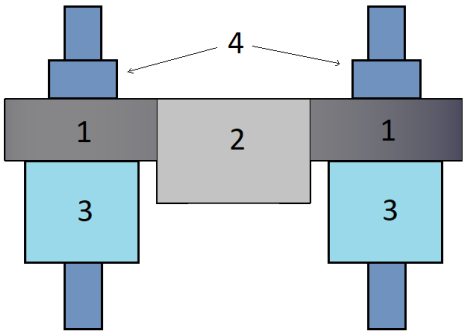


Figure XXX: Side view of top platform, divided into regions.

Region	Mass, m_i [g]	Position of centre of mass from the top surface, x_i [mm]
1	$\rho(10 \times \pi \times 25^2)$	5.0
2	$\rho(50 \times 25 \times t)$	$t/2$
3	2×190	19.5
4	2×60	10.0

Table XXX: Mass and position of centre of mass of each region (Density ρ=1.04×10^{-3}).

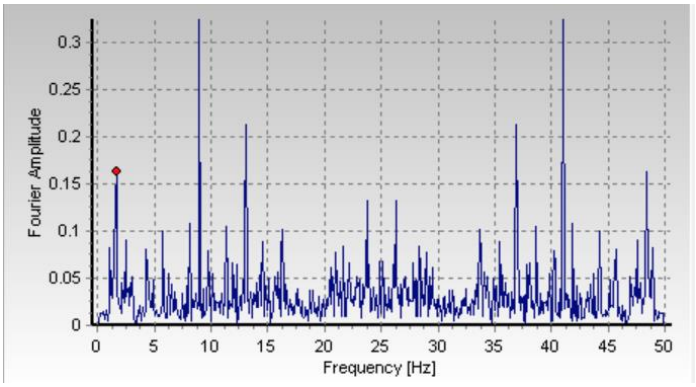
$$\bar{x} = \frac{\sum m_i x_i}{\sum m_i} = t$$

$$(625\rho)t^2 + (6250\pi\rho + 500)\mathrm{t} + (-8610 - 31250\pi\rho) = 0$$

$$t = 16.4mm \approx 17mm$$

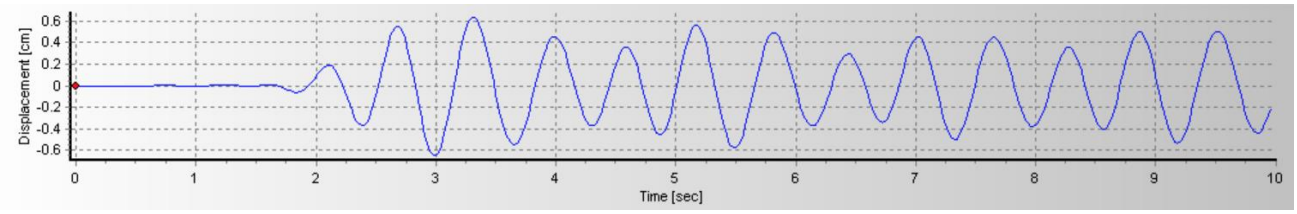
Appendix B: SeismoSignal analysis

(i) Test 1:



Test 1: Fourier Spectra without bandpass filter

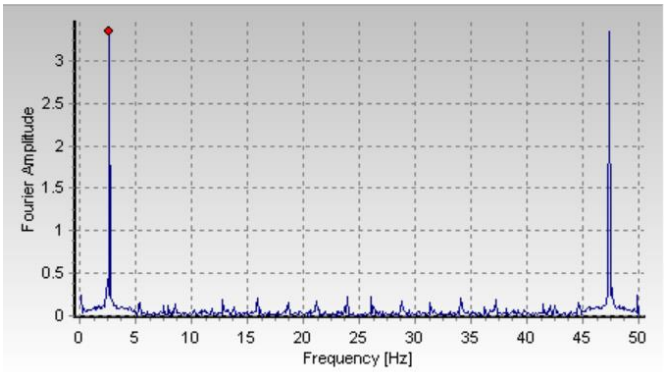
Distinct peak displacement		
Frequency [Hz]	Fourier Amplitude	Fourier Phase
1.66016	0.16393*	1.09954



Test 1: Displacement-Time graph with bandpass filter

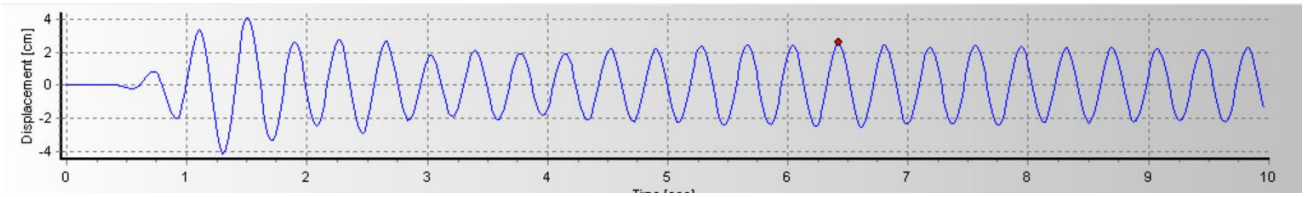
Bandpass filter		Umax [cm]
Freq 1	Freq 2	0.64731**
1.16	2.16	

(ii) Test 2:



Test 2: Fourier Spectra without bandpass filter

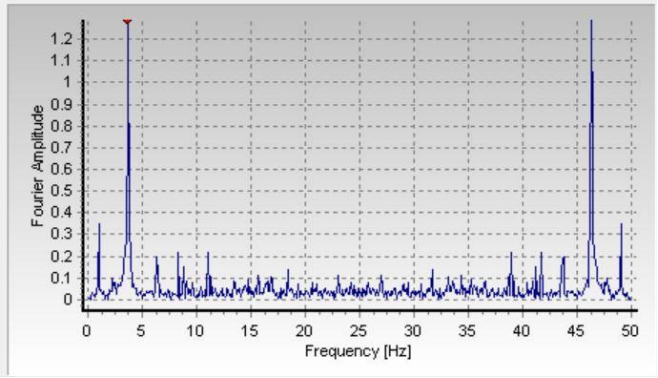
Distinct peak displacement		
Frequency [Hz]	Fourier Amplitude	Fourier Phase
2.63672	3.34594	-0.55191



Test 2: Displacement-Time graph with bandpass filter.

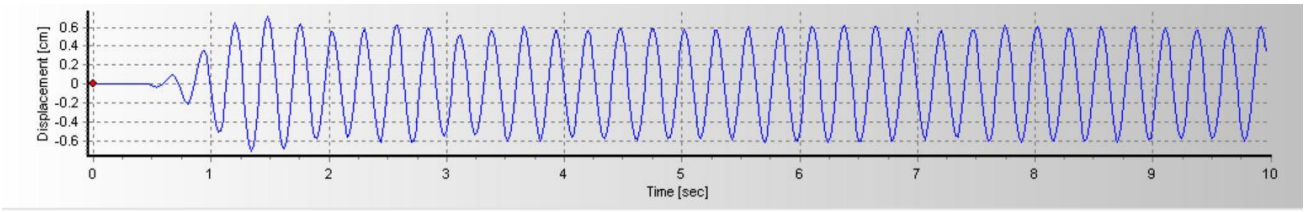
Bandpass filter		Umax [cm]
Freq 1	Freq 2	
1.60	3.60	2.57141**

(iii) Test 3:



Test 3: Fourier Spectra without bandpass filter

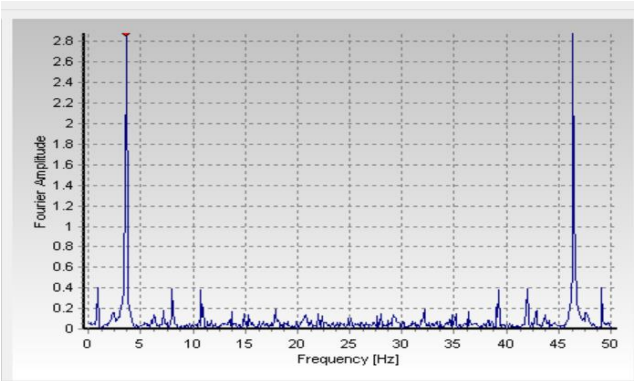
Distinct peak displacement		
Frequency	Fourier Amplitude	Fourier phase
3.71094	1.28661	0.83306



Test 3: Displacement-Time graph with bandpass filter.

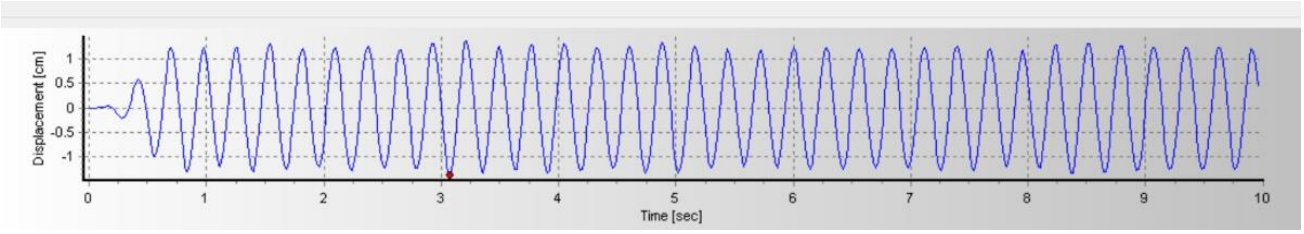
Bandpass filter		Umax [cm]
Freq 1	Freq 2	
2.70	4.70	0.62343**

(iv) Test 4:



Test 4: Fourier Spectra without bandpass filter

Distinct peak displacement		
Frequency	Fourier Amplitude	Fourier phase
3.61328	2.87897	0.43776



Test 4: Displacement-Time Graph with bandpass filter.

Bandpass filter		Umax [cm]
Freq 1	Freq 2	
2.60	4.60	1.38210**

Note:

*In test 1, the distinct peak displacement was taken at the frequency of 1.66016 Hz instead of the highest peak at frequency of 8.98438 Hz. This is because it gives a more reasonable value of maximum displacement as observed from the videos provided. There were also signs of the structure displacing with at least 2 dofs.

**In all tests, the maximum displacement, Umax, is taken as the maximum displacement of steady-state. (i.e. neglecting the transient state, which is the first 2 seconds of the test as seen in the Displacement-Time graph above.)

Appendix C: MATLAB script

```
% Dynamics Coursework

close all;
clear;
clc;

% Define constants (all units in m, N, kg)
Tnlim = [0.35,0.7];
ulim = 0.03;
tensilelim = [15,24.1]*(10^6);
shearlim = tensilelim/sqrt(3);
E = [1.07,2.03]*(10^9);
rho = [1.04,1.23]*(10^3);
psi = 0;
n = 2;

% Define constants and calculate beta, f0 and w
x = [0.01,0.01,0.02];
f = [1.5,2.5,3.5];
w = 2*pi*f;
f0 = x.*(w.^2);

%% Initiate & Iteration to determine Tn, umax and maxTensileStress

% Create vectors to iterate through
db = 0.001;
dd = 0.001;
dh = 0.001;
b = 0.015:db:0.042;
d = 0.002:dd:0.003;
h = 0.133:dh:0.133;

% Initiate matrices to store Tn, umax, tensileStress, mtotal
Tn = NaN(length(b),length(b),length(d),length(d),length(h),length(E));
umax = Tn;
maxTensileStress = Tn;
maxShearStress = Tn;
buckling = Tn;
sumd = 0;
sumb = 0;
for i = length(d):-1:1
    sumd = sumd+i;
end
for i = length(b):-1:length(b)-10
    sumb = sumb+i;
end
iterationNumber = sumb*sumd*length(h)*length(E);

l = 0;
for i1 = 1:11
    for i2 = i1:length(b)
        for i3 = 1:length(d)
            for i4 = i3:length(d)
                for i5 = 1:length(h)
                    for i6 = 1:length(E)
                        % Show progress
                        clc;
                        l = l+1;
                        fprintf('PART 1 ---
Loading: %.2f/100\n',l/iterationNumber*100);

                        % Calculate the natural period
```

```
                        fA = @(x)((b(i1)+(b(i2)-b(i1))/h(i5)*x).*(d(i3)+(d(i4)-
d(i3))/h(i5)*x));
                        m = (n*0.25)+(pi*(0.025^2)*0.01+0.05*0.025*0.017-
0.01*2*pi*(0.007^2)/4)*rho(i6)/2+(integral(fA,0,h(i5)/2))*rho(i6);
                        L = h(i5);
                        fK = @(x)(((L-x).^2)./((b(i2)-(b(i2)-
b(i1))/h(i5)*x).*((d(i4)-(d(i4)-d(i3))/h(i5)*x).^3)/12));
                        K = E(i6)/integral(fK,0,L);
                        wn = sqrt(K/m);
                        Tn(i1,i2,i3,i4,i5,i6) = 2*pi/wn;

                        % Continue if Tn does not meet criteria
                        if Tn(i1,i2,i3,i4,i5,i6)>=Tnlim(2) &&
Tn(i1,i2,i3,i4,i5,i6)<=Tnlim(1)
                            continue;
                        end

                        % Calculate umax for all 3 cases
                        beta = w./wn;
                        ust = f0./(wn.^2);
                        R = 1./sqrt(((1-beta.^2).^2)+((2*psi*beta).^2));
                        umax1 = ust.*R;
                        umax(i1,i2,i3,i4,i5,i6) = max(umax1);

                        % Continue if umax is larger than ulim
                        if umax(i1,i2,i3,i4,i5,i6)>=ulim
                            continue;
                        end

                        % Calculate the maximum tensile stress
                        Vmax = K*max([ust,umax(i1,i2,i3,i4,i5,i6)]);
                        fy = @(x)((d(i4)-(d(i4)-d(i3))/h(i5)*x)/2);
                        fI = @(x)((b(i2)-(b(i2)-b(i1))/h(i5)*x).*((d(i4)-(d(i4)-
d(i3))/h(i5)*x).^3)/12);
                        fM = @(x)(Vmax*(L-x));
                        x1 = 0:0.001:L;
                        tensileStress = fM(x1).*fy(x1)./fI(x1);
                        maxTensileStress(i1,i2,i3,i4,i5,i6) = max(tensileStress);

                        % Calculate the maximum shear stress
                        maxShearStress(i1,i2,i3,i4,i5,i6) = Vmax/fA(0);

                        % Calculate the weight
                        f1 = @(x)(((b(i1)+(b(i2)-b(i1))/h(i5)*x).*((d(i3)+(d(i4)-
d(i3))/h(i5)*x).^3)/12).*((sin(pi*x/h(i5)/2)).^2));
                        f2 = @(x)((cos(pi*x/h(i5)/2)).^2);
                        P =
E(i6)*(pi^2)/(h(i5)^2)*integral(f1,0,h(i5))/integral(f2,0,h(i5))/4;
                        if (m*9.81)>=P
                            buckling(i1,i2,i3,i4,i5,i6) = 0;
                        else
                            buckling(i1,i2,i3,i4,i5,i6) = 1;
                        end
                    end
                end
            end
        end
    end
end

%%

err = 0.0001;
merr = 0.001;
```

```
tensileFOS1 = 1.2;
shearFOS1 = 1;
tensileFOS2 = 1.2;
shearFOS2 = 1;
mtotal = NaN(length(b),length(b),length(d),length(d),length(h),length(E));

l = 0;
for i1 = 1:16
    for i2 = i1:length(b)
        for i3 = 1:length(d)
            for i4 = i3:length(d)
                for i5 = 1:length(h)
                    for i6 = 1:length(E)
                        % Show progress
                        clc;
                        l = l+1;
                        fprintf('PART 2 ---
Loading: %.2f/100\n',l/iterationNumber*100);

                        % Check if the chosen dimensions without error meet
                        % the criteria
                        if Tn(i1,i2,i3,i4,i5,i6)<Tnlim(2) &&
Tn(i1,i2,i3,i4,i5,i6)>Tnlim(1) && umax(i1,i2,i3,i4,i5,i6)<ulim &&
tensileFOS1*maxTensileStress(i1,i2,i3,i4,i5,i6)<tensilelim(i6) &&
shearFOS1*maxShearStress(i1,i2,i3,i4,i5,i6)<shearlim(i6) &&
buckling(i1,i2,i3,i4,i5,i6)==1
                            if err==0
                                % Calculate the total mass
                                fA = @(x)((b(i1)+(b(i2)-
b(i1))/h(i5)*x).*(d(i3)+(d(i4)-d(i3))/h(i5)*x));
                                mtotal(i1,i2,i3,i4,i5,i6) =
(pi*(0.025^2)*0.01+0.05*0.025*0.017-0.01*2*pi*(0.007^2)/4+0.01*pi*(0.055^2)-
4*0.01*pi*(0.01^2)/4)*rho(i6)/2+integral(fA,0,h(i5))*rho(i6);
                            else
                                % Initiate empty vectors
                                TnC = zeros(1,1+2^5);
                                umaxC = TnC;
                                maxTensileStressC = TnC;
                                maxShearStressC = TnC;
                                bucklingC = TnC;
                                TnC(1) = Tn(i1,i2,i3,i4,i5,i6);
                                umaxC(1) = umax(i1,i2,i3,i4,i5,i6);
                                maxTensileStressC(1) =
maxTensileStress(i1,i2,i3,i4,i5,i6);
                                maxShearStressC(1) =
maxShearStress(i1,i2,i3,i4,i5,i6);
                                bucklingC(1) = buckling(i1,i2,i3,i4,i5,i6);

                                % Calculate Tn, umax and maxTensileStress
                                % considering error
                                k = 1;
                                for j1 = -1:2:1
                                    for j2 = -1:2:1
                                        for j3 = -1:2:1
                                            for j4 = -1:2:1
                                                for j5 = -1:2:1
                                                    k = k+1;

                                                    % Calculate the natural period
                                                    fA =
@(x)(((b(i1)+j1*err)+((b(i2)+j2*err)-
(b(i1)+j1*err))/(h(i5)+j5*err)*x).*((d(i3)+j3*err)+((d(i4)+j4*err)-
(d(i3)+j3*err))/(h(i5)+j5*err)*x));
```

```
                                                    m =
(n*0.25)+(pi*(0.025^2)*0.01+0.05*0.025*0.017-
0.01*2*pi*(0.007^2)/4)*rho(i6)/2+(integral(fA,0,(h(i5)+j5*err)/2))*rho(i6);
                                                    L = h(i5)+j5*err;
                                                    fK = @(x)(((L-
x).^2)./(((b(i2)+j2*err)-((b(i2)+j2*err)-
(b(i1)+j1*err))/(h(i5)+j5*err)*x).*(((d(i4)+j4*err)-((d(i4)+j4*err)-
(d(i3)+j3*err))/(h(i5)+j5*err)*x).^3)/12));
                                                    K = E(i6)/integral(fK,0,L);
                                                    wn = sqrt(K/m);
                                                    TnC(k) = 2*pi/wn;

                                                    % Calculate umax for all 3 cases
                                                    beta = w./wn;
                                                    ust = f0./(wn.^2);
                                                    R = 1./sqrt(((1-
beta.^2).^2)+((2*psi*beta).^2));
                                                    umax1 = ust.*R;
                                                    umaxC(k) = max(umax1);


                                                    % Calculate the maximum tensile
stress
                                                    Vmax = K*max([ust,umaxC(k)]);
                                                    fy = @(x)(((d(i4)+j4*err)-
((d(i4)+j4*err)-(d(i3)+j3*err))/(h(i5)+j5*err)*x)/2);
                                                    fI = @(x)(((b(i2)+j2*err)-
((b(i2)+j2*err)-(b(i1)+j1*err))/(h(i5)+j5*err)*x).*(((d(i4)+j4*err)-((d(i4)+j4*err)-
(d(i3)+j3*err))/(h(i5)+j5*err)*x).^3)/12);
                                                    fM = @(x)(Vmax*(L-x));
                                                    x1 = 0:0.001:L;
                                                    tensileStress =
fM(x1).*fy(x1)./fI(x1);
                                                    maxTensileStressC(k) =
max(tensileStress);


                                                    % Calculate the maximum shear
stress
                                                    maxShearStressC(k) = Vmax/fA(0);


                                                    % Calculate the weight
                                                    f1 =
@(x)((((b(i1)+j1*err)+((b(i2)+j2*err)-
(b(i1)+j1*err))/(h(i5)+j5*err)*x).*(((d(i3)+j3*err)+((d(i4)+j4*err)-
(d(i3)+j3*err))/(h(i5)+j5*err)*x).^3)/12).*((sin(pi*x/(h(i5)+j5*err)/2)).^2));
                                                    f2 =
@(x)((cos(pi*x/(h(i5)+j5*err)/2)).^2);
                                                    P =
E(i6)*(pi^2)/((h(i5)+j5*err)^2)*integral(f1,0,(h(i5)+j5*err))/integral(f2,0,(h(i5)+j
5*err))/4;
                                                    if (m*9.81)>=P
                                                        bucklingC(k) = 0;
                                                    else
                                                        bucklingC(k) = 1;
                                                    end
                                                end
                                            end
                                        end
                                    end
                                end

                                if all(TnC<Tnlim(2)) && all(TnC>Tnlim(1)) &&
all(umaxC<ulim) && all(tensileFOS2*maxTensileStressC<tensilelim(i6)) &&
all(shearFOS2*maxShearStressC<shearlim(i6)) && all(bucklingC)
                                    % Calculate the total mass
```

```
                                    fA = @(x)((b(i1)+(b(i2)-
b(i1))/h(i5)*x).*(d(i3)+(d(i4)-d(i3))/h(i5)*x));
                                    mtotal(i1,i2,i3,i4,i5,i6) =
(pi*(0.025^2)*0.01+0.05*0.025*0.017-0.01*2*pi*(0.007^2)/4+0.01*pi*(0.055^2)-
4*0.01*pi*(0.01^2)/4)*rho(i6)/2+integral(fA,0,h(i5))*rho(i6);
                                end
                            end
                        end
                    end
                end
            end
        end
    end
end
fprintf('Number of solutions = %i\n',nnz(~isnan(mtotal)));

%% Find the dimensions of the structure with the minimum mass

% Determine the index to the minimum mass
[A1,i6] = min(mtotal,[],6);
[A2,i5] = min(A1,[],5);
[A3,i4] = min(A2,[],4);
[A4,i3] = min(A3,[],3);
[A5,i2] = min(A4,[],2);
[mmin,i1] = min(A5,[],1);
i2 = i2(i1);
i3 = i3(i1,i2);
i4 = i4(i1,i2,i3);
i5 = i5(i1,i2,i3,i4);
i6 = i6(i1,i2,i3,i4,i5);
fprintf('The minimum mass is %fkg: b1=%.3fm b2=%.3fm d1=%.3fm d2=%.3fm h=%.3fm
material=%i n=%i\n',mmin,b(i1),b(i2),d(i3),d(i4),h(i5),i6,n);

% Calculate the natural period
fA = @(x)((b(i1)+(b(i2)-b(i1))/h(i5)*x).*(d(i3)+(d(i4)-d(i3))/h(i5)*x));
m = (n*0.25)+(pi*(0.025^2)*0.01+0.05*0.025*0.017-
0.01*2*pi*(0.007^2)/4)*rho(i6)/2+(integral(fA,0,h(i5)/2))*rho(i6);
L = h(i5);
fK = @(x)(((L-x).^2)./((b(i2)-(b(i2)-b(i1))/h(i5)*x).*((d(i4)-(d(i4)-
d(i3))/h(i5)*x).^3)/12));
K = E(i6)/integral(fK,0,L);
wn = sqrt(K/m);
TnS = 2*pi/wn;
fprintf('The natural period is %.4fs. \n',TnS);

% Calculate umax for all 3 cases
beta = w./wn;
ust = f0./(wn.^2);
R = 1./sqrt(((1-beta.^2).^2)+((2*psi*beta).^2));
umax1 = ust.*R;
umaxS = max(umax1);
fprintf('The maximum dynamic displacement for load case %i is %.6f.
\n',[[1,2,3];umax1]);
fprintf('The maximum displacement is %.6f. \n',max([ust,umaxS]));

% Calculate the maximum tensile stress
Vmax = K*max([ust,umaxS]);
fy = @(x)((d(i4)-(d(i4)-d(i3))/h(i5)*x)/2);
fI = @(x)((b(i2)-(b(i2)-b(i1))/h(i5)*x).*((d(i4)-(d(i4)-d(i3))/h(i5)*x).^3)/12);
fM = @(x)(Vmax*(L-x));
x1 = 0:0.001:L;
tensileStress = fM(x1).*fy(x1)./fI(x1);
maxTensileStressS = max(tensileStress);
fprintf('The maximum tensile stress is %.2f. \n',maxTensileStressS);
```

```
tensileFOS = tensilelim(i6)/maxTensileStressS;
fprintf('The factor of safety for tensile strength is %.3f. \n',tensileFOS);

% Calculate the maximum shear stress
maxShearStressS = Vmax/fA(0);
fprintf('The maximum shear stress is %.2f. \n',maxShearStressS);
shearFOS = shearlim(i6)/maxShearStressS;
fprintf('The factor of safety for shear strength is %.3f. \n',shearFOS);

% Calculate buckling load
f1 = @(x)(((b(i1)+(b(i2)-b(i1))/h(i5)*x).*((d(i3)+(d(i4)-
d(i3))/h(i5)*x).^3)/12).*((sin(pi*x/h(i5)/2)).^2));
f2 = @(x)((cos(pi*x/h(i5)/2)).^2);
P = E(i6)*(pi^2)/(h(i5)^2)*integral(f1,0,h(i5))/integral(f2,0,h(i5))/4;
fprintf('The weight is %.2f and the buckling load is %.2f. \n',m*9.81,P);
```